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Human Cerebellar Organoids with functional Purkinje Cells

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SUMMARY:

Research on human cerebellar development and disease has been hampered by the need for a human cell-based system that recapitulates the human cerebellum's cellular diversity and functional features. Here, we report a human organoid model (hCerOs) capable of developing the complex cellular diversity of the fetal cerebellum, including human-specific rhombic lip progenitor populations that have never been generated *in vitro* prior to this study. Two months old hCerOs form distinct cytoarchitectural features, including laminar organized layering and create functional connections between inhibitory and excitatory neurons that display coordinated network

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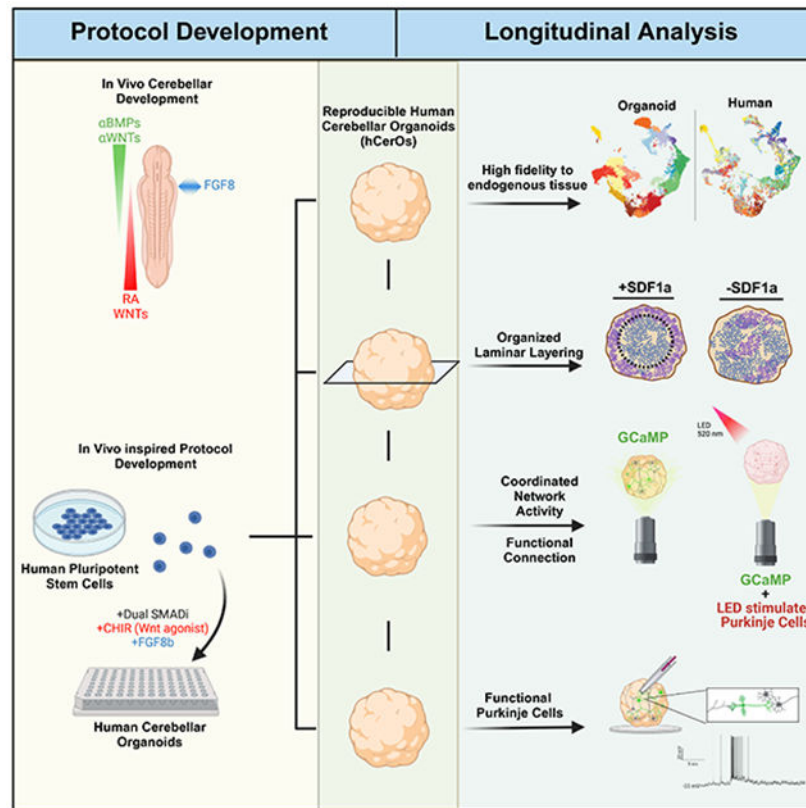
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activity. Long-term culture of hCerOs allows for healthy survival and maturation of Purkinje cells that display molecular and electrophysiological hallmarks of their *in vivo* counterparts, addressing a long-standing challenge in the field. This study therefore provides a physiologically relevant, all-human model system to elucidate the cell type specific mechanisms governing cerebellar development and disease.

Graphical Abstract



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We developed (hCerOs) a human organoid model replicating the fetal cerebellum's cellular diversity including the generation of human-specific rhombic lip progenitors recently linked to medulloblastoma initiation. hCerOs can transiently form laminar organized layering and develop functional neurons, including Purkinje cells, providing an all-human, physiologically relevant model system for cerebellar research.

Keywords

human cerebellum; human pluripotent stem cell-derived organoids; single-cell RNA seq; human-specific rhombic lip subventricular zone; functional Purkinje cells; network activity; spatial transcriptomics

INTRODUCTION

The cerebellum has been gaining substantial attention, given emerging evidence of its role in cognitive functions, including language, spatial processing, working memory, executive functions, and emotional processing, in addition to its well-described role in motor behaviors¹⁻³. In all mammals, the cerebellum is initially derived from the alar lamina (dorsal) rhombomere 1 (r1) region caudal to the isthmus organizer at the midbrain-hindbrain boundary. The r1 comprises two distinct progenitor zones including the PTF1A+ ventricular zone (VZ) and the ATOH1+ Rhombic Lip (RL). The VZ produces GABAergic neurons, including Purkinje Cells, molecular layer interneurons (MLI), and deep cerebellar nuclei (iCN), while the more dorsally located RL gives rise to glutamatergic neurons, including granule cells (GC), unipolar brush cells (UBC), and large deep cerebellar nuclei projection neurons (eCN)⁴⁻⁹.

Despite the conservation of early developmental stages of cerebellar ontogenesis across species, human-specific traits, including altered neuronal subtype ratios and increased folial complexity, have been identified when compared to non-human mammalian counterparts^{1,10}. In addition, comparison between human and non-human primates has identified the presence of a human-specific population of neural progenitors derived from the rhombic lip¹¹. This is important, as abnormal development of the human rhombic lip is associated with multiple cerebellar disorders including Dandy-Walker malformation, cerebellar vermis hypoplasia, and medulloblastoma, the most common metastatic childhood brain tumor¹². Other cerebellar disorders including ASD and ataxia have been associated with the degeneration of Purkinje cells, the main output neurons of the cerebellar cortex¹³⁻¹⁵. These disorders have mainly been studied in mouse models, which cannot fully recapitulate human disease phenotypes. An example of this includes the inability of mice to recapitulate the Purkinje cell loss phenotype observed in patients with ataxia telangiectasia^{16,17}. Thus, there is a pressing need to develop an all-human cell-based culture system capable of generating functional Purkinje cells.

Here we report an organoid model that allows healthy long-term survival and maturation of functional cerebellar cells, including Purkinje neurons, in an all-human 3D context addressing a longstanding challenge of the field. We show that cerebellar organoids reliably generate spatially segregated ventricular and rhombic lip progenitor zones, which give rise to the cellular diversity of the human cerebellum within and across multiple cell lines. At two months in culture, we observed the emergence of an organized laminar layering with inhibitory and excitatory cerebellar neuronal subtypes, which are functionally connected and display coordinated activity, which increases over a period of six months in culture. Finally, long-term culture of cerebellar organoids allows the maturation of Purkinje cells that display the transcriptomic profile and electrophysiological features of functional neurons.

RESULTS

Human cerebellar organoids (hCerO) reproducibly generate the cellular diversity of the human cerebellum.

The cerebellum develops from the most rostral region of the rhombomere 1, an area of the developing neural tube just caudal to the isthmus organizing center, which defines the midbrain-hindbrain boundary. Modulation of key morphogen pathways has allowed for establishing protocols for the differentiation of 3D cultures of human pluripotent stem cells into organoids specific to particular brain regions including the cerebellum. However, it has been shown that caudalization relying on only FGF2+insulin, commonly used to generate cerebellar organoids¹⁸⁻²¹ is inferior to other patterning strategies in generating a hindbrain fate²². This is not surprising, as foundational neurodevelopmental biology studies have shown that the combinatorial activity of several caudalizing factors is required *in vivo* to overcome the preference of neural tissue to develop an anterior character²³. To develop a reproducible *in vitro* model of the cerebellum we established an organoid protocol based on the modulation of the signaling molecules required for the development of the isthmus organizer *in vivo*. Previous robust *in vitro* generation of midbrain/hindbrain fates has incorporated the activation of WNT and FGF8 signaling²⁴⁻²⁷. Therefore, to promote neuroectodermal differentiation toward a cerebellar fate, cell aggregates were simultaneously treated with the dual SMAD inhibitors Noggin and SB431542²⁸ and the canonical WNT pathway activator CHIR-99021. On day 4, FGF8b was then added to specify the midbrain-hindbrain boundary and suppress midbrain fate, re-creating the isthmus organizing center seen *in vivo* (Figures 1A and 1B)²⁹⁻³².

The use of this patterning strategy along with the use of insulin and growth factor reduced media (gfCDM)^{18,33} led to the successful elimination of forebrain progenitors, while the commonly used FGF2+insulin patterning strategy, in our hands, yielded high levels of forebrain as observed through the use of a forebrain reporter cell line (PGP::tdTOMATO::FOXG1) (Figures S1A-S1F, S1H, S1I, S1K). Our patterning strategy with insulin and gfCDM produced a "rhombic lip" organoid that only expressed markers of the excitatory lineage (ATOH1 and BARHL1). This organoid lacked ventricular zone progenitors marked by the inhibitory lineage marker KIRREL2+ and postmitotic Purkinje cell precursor derivatives marked by SKOR2 (Figure S1K). The ability to generate BARHL1+ cells was also seen in the previously published FGF2+insulin strategy in gfCDM, however, a significant level of forebrain was also detected (Figure S1I). It is known that base media can affect regional patterning³³. We thus created a cerebellar induction strategy based on an optimized combination of base media to lock in the midbrain/hindbrain boundary identity and to generate a pure compendium of cerebellar cell types (Figures S1J, S1L, S1M-S1R). With this protocol, the hCerOs properly caudalized with increased expression of caudal markers such as IRX3 and GBX2 (Figure 1D; Figure S1G). By day 30, the hCerOs reliably generated spatially segregated ventricular (KIRREL2) and rhombic lip (ATOH1) progenitor zones (Figure 1C; Figures S1S, S1T), as well as their derivatives: excitatory BARHL1+ granule cell progenitors and newborn SKOR2+ inhibitory Purkinje neurons (Figure S1L). Importantly, the ability to generate these two populations of neuronal progenitors was identified across three hiPSC lines including PGP1, 11a, and D2, and one

hESC line, H9 (Figures S2A-S2D). Despite the H9 line having a tendency to favor dorsal forebrain development when instructed to become ectoderm, it was still able to generate SKOR2⁺ Purkinje cell progenitors, although it was not consistent in producing the more dorsal granule cell population (Figure S2E). These data indicated the potential of our protocol to generate all the cell types of the developing cerebellum across multiple hPSC lines.

We conducted single-cell RNA sequencing on 14,947 cells from 3 individual organoids that were 2 months old. Through VoxHunt, a tool used to assess cell type composition and developmental stage of neural organoids³⁴, we mapped the 2-month-old hCerO's scRNA-seq data onto the BrainSpan human transcriptomic dataset and found that our dataset the highest scaled correlation with cerebellum (CB) when compared to primary brain samples between 12-24 PCW (Figure 1E; Figure S3A). These findings have also been recently corroborated by an independent analysis, contributing to the construction of a human brain organoid atlas³⁵, which was benchmarked against the reference dataset from developing human brains³⁶. In order to understand the diversity of cells present in our hCerOs, we systematically categorized the clusters by comparing signatures of differentially expressed genes to a pre-existing single cell dataset of endogenous cell types of the human developing cerebellum³⁷ (Figure 1F and 1G; Figure S3B-S3J). Uniform Manifold Approximation and Projection (UMAP) dimensionality reduction identified 16 major clusters, including the two main progenitor types, the *KIRREL2*⁺, *PTF1A*⁺, and *VIM*⁺ ventricular zone progenitors, and *ATOH1*⁺ and *BARHL1*⁺ rhombic lip progenitors, which derive the inhibitory and excitatory neuronal populations. *In vivo*, the ventricular zone also gives rise to Bergmann glia, a scaffolding cell type necessary to aid in the migration of maturing granule cells into their proper orientation³⁸. Indeed, within our datasets, we were also able to identify a cluster of cells that expressed key Bergmann glial markers including *PTPRZ1*, *EDNRB*, *GFAP*, *HOPX*, *SLCA4A4*, and *EDNRB*. Among the neuronal clusters, we identified cells expressing *in vivo* canonical markers of Purkinje cells, including *SKOR2*, *RORA*, *FOXP2*, and *CALB1*. In addition, we classified three other GABAergic (*STMN2*⁺, *GAD2*⁺) clusters: interneuron precursors expressing *PAX2*, molecular layer interneurons expressing *SOX14* and *DMBX1*, and inhibitory deep cerebellar nuclei with markers *MEIS2*, *LHX9*, and *IRX3*. Additionally, the Purkinje cells and cerebellar nuclei had corresponding “immature” clusters, which had a similar transcriptomic profile as their more mature counterparts but lacked the expression of genes associated with later stages of maturation *in vivo*. These include *CALB1*, *FOXP2*, and *RORA* for Purkinje cells, and *NEUROD1/2* for cerebellar nuclei. We also identified three types of glutamatergic neurons (*STMN2*⁺, *SLC17A7*⁺) expressing Unipolar Brush Cell (UBC) marker *EOMES*, Granule Cell progenitor (GCP) markers *ATOH1* and *BARHL1*, and mature Granule Cell (GC) expressing higher levels of more mature neuronal markers including *NEUROD1* and *NHLH1*. We conducted a detailed analysis of the expression level of the top 100 genes that characterize human Purkinje cells³⁷ and compared it with the Purkinje cells present in our 2-month-old hCerOs. Our findings revealed that the Purkinje cells in our hCerOs expressed a considerable number of these markers that define Purkinje cells, which are observed between 9-20 PCW of the developing human cerebellum (Figure 1 in Mendeley Data).

Previously reported work has revealed that the human rhombic lip is divided into a ventricular and subventricular zone (RLvz and RLsvz), a feature that is not shared by any other vertebrates^{11,39}. Because of the human-specific nature of our system, we then sought to identify these uniquely human rhombic lip progenitor subsets. To determine if the cell types specific to the RLvz and RLsvz are present in our system, we subclustered the RL and dividing VZ clusters, which were then integrated with 1018 cells identified to have RL identity in the human developing cerebellum dataset³⁷ (Figure 2A). We observed that our cells correlated well with the RLvz and RLsvz clusters of the human dataset, indicating that we have both populations of progenitors present in our organoids (Figure 2B-2D). We then took the DEGs from the human dataset used to characterize the RLvz, RLsvz, and the IZ. This comparison confirmed the transcriptomic similarity of hCerO-derived cells to distinct sub-clusters found in the human dataset (Figure 2E-2G). We also saw the expression of classic RL markers, including *MKI67*, *PAX6*, and *LMX1A*, throughout our dataset, as was seen in the human dataset. Additionally, the expression of *WLS*, *SOX2*, and *CRAYAB* was predominantly expressed in the RLvz cluster, while higher levels of expression for *EOMES*, *DPYD*, *GPC6*, *OTX2*, and *BARHL1* were seen in the RLsvz (Figure 2H).

In addition to the cell types deriving directly from the two progenitor zones of the cerebellum, we also observed the ability of hCerOs to co-develop other cell types also found in the human fetal dataset known to be important contributors to cerebellar development *in vivo*^{40,41}. These cell types include *TTR*⁺ Choroid Plexus (*TOP2A*⁺ dividing choroid plexus), *LUM*⁺*DCN*⁺ Meninges, and *GDF7*⁺ Roof Plate cells.

Then, we utilized this single-cell data set to look at the expression of risk genes linked with a variety of disorders associated with cerebellar dysfunction, including intellectual disability (ID), autism spectrum disorder (ASD), spinocerebellar ataxia (SCA), Joubert Syndrome, Alzheimer's disease, and cerebellar malformations to assess cell type enrichment in 2-month-old cerebellar organoids (Figure 2 **in Mendeley Data and Supplementary Text in Mendeley Data**) and found an overall enrichment in neuronal cell types.

Finally, to assess organoid-to-organoid reproducibility in cell type composition, we performed scRNA-seq on three 2-month-old hCerOs (org1: 4,817 cells, org2: 6,768 cells, org3: 3,362 cells) and found that these organoids can generate all the main cell types of the developing cerebellum. (Figure S3J).

Altogether, our data show that our hCerO protocol reproducibly generates all the major cell types of the developing cerebellum, including human-specific progenitor subtypes of the fetal rhombic lip and Purkinje cells that express the transcriptional profile of their *in vivo* counterparts. Thus, this model comprehensively allows for the examination of cell type-specific mechanisms governing cerebellar development and disease.

hCerOs display organized laminar layering reminiscent of the EGL and PCL of the developing cerebellar anlage.

While analysis of the cell type composition of hCerOs demonstrated that the generation of cerebellar cellular diversity is an intrinsic process that does not require the *in vivo* cues of the developing embryo, we next investigated whether distinctive cytoarchitectural features

could also organically emerge from this system. During the second trimester of human cerebellar ontogenesis, the excitatory granule cell precursors undergo directed tangential migration along the pial surface to form the external granule layer (EGL) (Figure 3A)⁴², meanwhile, the Purkinje cells migrate radially toward the pial surface, settle below the EGL, and form the multicell Purkinje layer (PCL). Despite the presence of these neuronal populations, we found that this mechanism is not intrinsically regulated in 2-month-old hCerOs, a stage roughly equivalent to the second trimester of human fetal development. Thus, the 2-month-old hCerOs are unable to organize as seen *in vivo*. Therefore, we decided to instruct their migration by using external cues.

During fetal development, SDF1a (also known as CXCL12), a chemoattractant released by meningeal cells, guides the tangential migration of granule cell progenitors along the pial surface through the activation of the corresponding CXCR4 receptor⁴³. *In vivo* CXCR4 is expressed in granule cell progenitors⁴⁴, and as these cells mature to granule cells, they lose expression of the CXCR4 receptor and consequently sensitivity to SDF1a and migrate radially away from the pial surface⁴⁵. In agreement with *in vivo* expression patterns, we found that BARHL1+ granule cell progenitors express CXCR4 in our 2-month-old hCerOs (Figure 3B and 3C). It is worthwhile to mention that CXCR4 is also found in neurogenic progenitors and cerebellar ventricular radial glia *in vivo*⁴⁶⁻⁴⁸ as is seen with the RNA expression of CXCR4 in our single cell data further corroborating the similarity between the expression pattern of CXCR4 *in vivo* and in hCerOs. To assess the response of hCerOs to SDF1a, we added 100ng/ml of recombinant SDF1a into our maintenance media every 3 days (CerDM3) for 1 month (between day 30-60). After treatment, we observed the BARHL1+ and PAX6+ cells aligned adjacent to the edge of the organoid, while the SKOR2+ cells appeared beneath that layer (Figure 3D-3I). When co-staining for the Purkinje cell zone (CALB1+) and the RL-derivative zone (BARHL1+), we noticed that there appear to be two distinct layers in the +SDF1a treated hCerOs (Figure 3J-3N), reminiscent of the apical-basal polarity seen in the early stages of murine cerebellar development (~E15). Interestingly, a previous attempt to modify the structural organization of cerebellar organoids using 300ng/mL SDF1a for 7 days following FGF19 treatment, yielded a reversed organization of the cerebellar anlage¹⁸, suggesting the importance of the fine-tuning the duration and concentration of chemoattractants to build specific cytoarchitectural features within organoid models. It should be noted that the effect of SDF1a on the layering of the laminar structure gradually diminished when SDF1 was discontinued at day 60. By the time the organoids reached the age of 6 months, we observed that granule cells and Purkinje cells were intermingled (as shown in Figure 3O), possibly due to the absence of external scaffolding and cues provided by other cell types *in vivo*. Altogether, we found that while laminar layering is not intrinsically encoded in developing hCerOs, administration of development-inspired external cues at the right concentration and timing, can aid in improving the ability of organoids to recapitulate features of the anatomical organization found in the developing human brain.

hCerOs display functionally mature network activity in long-term cultures, resembling patterns of in vivo cerebellar circuits.

Following the characterization of the cellular diversity and cytoarchitectural features of hCerOs, we interrogated whether spontaneously active networks were present. We performed live two-photon microscopy-based imaging of intact 2-month-old hCerOs, following infection with AAV8 GCAMP6f2 virus to record intracellular calcium dynamics (Figure 4A). By performing an unbiased analysis and categorization of single-cell calcium dynamics, we identified individual Tetrodotoxin (TTX, a sodium channel blocker) sensitive Ca^{2+} neuronal calcium signals within hCerOs, indicative of spontaneous neuronal activity (Figure 4B-4F, Figure S4G)^{49,50}. Importantly, we found that levels of spontaneous activity, as shown by the average activation per frame, amplitude, and duration of calcium events, were similar across organoids derived from 3 hiPSC lines: D2, PGP1, and 11a (Figure S4A-S4F; Figure 3 in Mendeley Data). This further indicates that hCerOs display a high degree of organoid-to-organoid reproducibility, not only in terms of cellular composition (Figure S3J), but also at a functional level.

In addition, the single cell calcium signals were then used to characterize brain organoid physiological activity at network levels. We performed clustering analysis and found the presence of multiple functional clusters spatially distributed across the entire organoid, suggesting that as early as 2 months, neurons in hCerOs display coordinated firing indicative of the emergence of spontaneous network activity (Figure 4G and 4H).

We then investigated whether long-term culture of hCerOs would lead to changes in network activity. Comparison of 2- and 6-month-old hCerO neuronal spiking profiles showed that 6-month-old organoids had an overall increase in average activation per frame and a decrease in interpeak time per neuron, indicating an increase in overall calcium transient activity (Figure 4I and 4J; Video S1 and S2) and more neurons contributing to the activation of organoids. Additionally, we see an increase in the average correlation coefficient (Figure 4K) and an increase in burstiness within clusters (Figure 4L), suggesting a transition from single mature neuronal events into spontaneous coordinated activity across multiple neurons.

As hCerOs contain a mixture of glutamatergic granule cells/eCN/UBC and GABAergic Purkinje cells/MLI/iCN/PIP cells, we sought to assess their distinct functional contributions through bath application of NMDA and PTX to 2- and 6-month-old organoids. We found that combined treatment of 6-month-old hCerOs with NMDA+PTX greatly increased burstiness in each cluster, compared to treatment with NMDA alone. This indicates a contribution of both glutamatergic and GABAergic neurons to the overall activity within clusters. The level of activity post-treatment with NMDA+PTX is significantly higher than 2-month-old hCerOs, indicating an increase in the level of functional connectivity in long-term cultured hCerOs (Figure 4M-4O). In addition, validation through immunostaining confirmed the presence of neurons expressing Calbindin and GABA, indicating that Purkinje neurons can release the GABA neurotransmitter (Figure S4H). We then assessed the active involvement of Purkinje cells in the network by infecting 6-month-old hCerOs with a lentiviral construct expressing the silencing opsin eOPN3 under the control of L7-PCP2 promoter⁵¹. PCP2 is selectively expressed in Purkinje cells and has been used previously to identify Purkinje cells in live human culture systems²⁷. Remarkably, LED stimulation

inhibited Purkinje cells, leading to elevated calcium activity levels in the organoid (Figure 4P-4R; Figure S4I-S4K, Video S3 and S4). These results strongly suggest that Purkinje cells actively contribute to the network's function.

Overall, this data shows the emergence of functional network activity in hCerOs after as little as 2 months in culture. Long-term culture increases the level of functional connectivity between excitatory and inhibitory neurons within hCerOs. The data provides a foundation for the use of human cerebellar organoids to model functional impairment associated with dysfunction in the development of cerebellar microcircuits.

Functionally mature human Purkinje neurons develop within long-term culture of hCerOs.

To our knowledge, functional maturation of Purkinje cells has never been obtained in an all-human culture system. Therefore, we focused our investigation on the functionality of these neurons, which are dysfunctional in an array of neurological disorders. We identified Purkinje cells in 6-month-old hCerOs through the expression of classical *in vivo* marker genes, such as PCP2, DAB1, RORA, and FOXP2, (Figure 5A-5C; Figure 4A **in Mendeley Data; Table 1 and Table 2 in Mendeley Data**), via spatial transcriptomics, which also gave us access to the spatial localization of these neurons (Figure 5D; Figure 4B **in Mendeley Data**). Having identified these neurons on the outer edge of the organoids, we decided to conduct whole cell patch clamp recordings on intact hCerOs to prevent the disruption of the complex neuronal connections.

Following the transduction of hCerOs with a virus that fluorescently reports for the expression of PCP2/L7, we performed whole-cell patch-clamp recordings of PCP2/L7+ neurons labeled with an AAV8.L7-6.eGFP.WPRE.hBG (Figure 5E), and found that PCP2/L7+ cells (n=19) at days 135–220 presented an average resting membrane potential (RMP) of –53.88 mV (**Table 3 in Mendeley Data**). 84.2% of the recorded cells showed the ability to fire multiple mature induced action potentials with an average amplitude of 68.80 mV (Figure 5F and 5H). 42.10% of the recorded neurons presented the ability to fire spontaneous TTX-sensitive action potentials with an average frequency of 14.63 Hz (Figure 5G and 5I; Figure 4C **in Mendeley Data**), highlighting the mature profile of the recorded neurons. Importantly, as observed in cerebellar Purkinje neurons *in vivo*, we were able to detect the presence of I_h current (Figure 5J and 5K) and repetitive spontaneous firing (Figure 5L), a profile distinctive of Purkinje cells^{52,53}, in our PCP2/L7+ cells.

We conducted imaging of the Calbindin marker on whole hCerO organoids that were 6 months old to determine the morphology of the Purkinje neurons. Our findings, which were in line with calcium imaging analysis and patch-clamp data, showed the existence of Calbindin-labeled Purkinje neurons and significant arborization and diverse morphology at the 6-month mark that is similar to a human Purkinje cell in fetal tissue between 22-28PCW⁵⁴ (Figure 5M). Previous research indicated that postnatal mice glia and granule cells are necessary for the functional maturation of hiPSC-derived Purkinje cells^{18,27}. However, our study demonstrates the development of functional Purkinje cells in an all-human system. These cells express the classical *in vivo* Purkinje cell markers and exhibit the distinct electrophysiological profile of their *in vivo* counterparts.

DISCUSSION

Our study establishes an organoid model to investigate the cellular interactions that orchestrate development, homeostasis, and diseases of the human cerebellum. This is enabled by the establishment of a protocol that reproducibly generates all the major cell and supporting cell types that aid in overall cerebellar development, including choroid plexus and roof plate cells. This demonstrates the existence of an intrinsic cellular program that regulates the acquisition of the molecular identity and functional characteristics of the diverse repertoire of human cerebellar neurons. This diverse compendium of cell types includes a human-specific progenitor subtype of the Rhombic Lip (RLsvz) population that has never been generated *in vitro* prior to this study.

The generation of RLsvz in our study signifies that hCerOs can serve as a valuable tool for advancing our understanding of cerebellar evo-devo biology and for exploring diseases associated with disruptions in rhombic lip development. These diseases include conditions such as Dandy-Walker Syndrome⁵⁵ and pediatric cancers^{56,57}.

Furthermore, MLIs play a crucial role in shaping the spike activity of Purkinje cells within cerebellar circuitry. Consequently, the presence of these cells within hCerOs, coupled with the consistent and high degree of cellular diversity in our experimental protocol, could establish an environment conducive to the functional maturation of Purkinje cells. Interestingly, our 6-month-old organoid-derived Purkinje cells resemble 22-28 weeks post-conception Purkinje cells from the fetal human cerebellum⁵⁴, suggesting a similar developmental trajectory *in vitro* as observed *in vivo*. This significant milestone, previously attainable only through the co-culture of mouse glia and granule cells^{18,24,27}, has been challenging to achieve.

Although Purkinje cells are primarily affected in numerous neurodegenerative disorders, it is increasingly clear that interactions with other cell types, including, but not limited to, glia cells, may have a central role in disease pathogenesis⁵⁸. This further underscores the importance of developing an all-human system.

As the link between cerebellar dysfunction and neuropsychiatric connectopathies, including ASD and ID, becomes more evident^{14,59}, the generation of multiregional organoids that can model long-range connections within the patients' brains becomes a priority. In this scenario, hCerOs with functional Purkinje cells, the main neuronal output of the cerebellum, will enable the recapitulation of cerebellar connectivity with other brain regions and accelerate therapeutic discovery.

LIMITATIONS OF THE STUDY

This study demonstrates that hCerOs have the ability to create ventricular and rhombic lip progenitor areas with spatial separation, resulting in excitatory BARHL1+ granule cell progenitors and newborn SKOR2+ inhibitory Purkinje neurons in multiple hPSC cell lines. It is important to note, however, that only one hPSC cell line has been subjected to extensive single-cell RNA sequencing and Vox Hunt analysis to determine the exact cellular diversity and regional identity of individual organoids. Therefore, we cannot rule

out that precise cellular composition may differ across cell lines, and that different cell lines may produce slightly different results and require adjustments in morphogen concentrations. Future studies using large-scale single-cell RNA sequencing analysis in a large number of hPSC cell lines comparing our protocol with previously published protocols would also be beneficial to understand the propensity of individual cell lines to selectively generate the cerebellar territory and prevent the formation of more rostral regions, which could further improve the effectiveness of the protocol. In our study, we have only carried out functional characterization of Purkinje neurons up to 6 months old organoids. It would be interesting to investigate the functional and morphological development of Purkinje cells over a longer period of developmental time. Finally, in future experiments, it would be exciting to integrate live cell imaging analysis to monitor cell migration in hCerOs. This would help to better dissect the tangential migration of granule cells around the edges of the organoids as well as the transient formation of organized laminar layering reminiscent of the developing cerebellar anlage under SDF1a treatment.

Star Methods:

RESOURCE AVAILABILITY

Lead contact: Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Giorgia Quadrato (quadrato@usc.edu).

Materials availability: This study did not generate new unique reagents.

Data and code availability:

- Single-cell RNA-seq data have been deposited at GEO and are publicly available as of the date of publication. Accession numbers are listed in the key resources table. Additional Data associated with Figures 1, 4, and 5 have been deposited in Mendeley Data and are publicly available as of the date of publication. DOIs are listed in the key resources table.
- All original code has been deposited at Github and is publicly available as of the date of publication. The DOI is listed in the key resources table.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Cell Lines: The PGP1 (Personal Genome Project 1) human iPS cell line was from the laboratory of G. Church; PGP1-FOXG1 human iPS cell lines were obtained from Harvard Stem Cell Institute; the H9 human ES cell line was purchased from WiCell, the 11a human iPS cell line was from the laboratory of Kevin Eggen, and the D2 human iPSC cell line was from the laboratory of Marcelo Coba. All stem cell lines were cultured on Geltrex (Gibco)-coated tissue culture plates, using mTeSR1 medium (Stem Cell Technologies) with 100 U/ml penicillin and 100 µg/ml streptomycin (Corning) at 37°C in 5% CO₂. All human stem cells were maintained below passage 50 and periodically karyotyped via the G-banding

Karyotype Service at Children's Hospital Los Angeles and were negative for mycoplasma (MycoAlert Plus Mycoplasma Detection Kits).

METHOD DETAILS

Organoid differentiation—For differentiation, feeder-free cultured human stem cells, 80% confluent, were dissociated to single cells using room temperature Accutase (Sigma) and were reaggregated in ultra-low-attachment, V-bottomed, 96-well plates (Sbio) at 6,000 cells per 100 μ l per well, in a growth factor reduced chemically defined medium (gfCDM), based on the Wataya ³³, consisting of 1:1 IMDM (Gibco) and F-12 (Gibco), 5mg/ml >99% Bovine Serum Albumin (BSA) (Sigma), 1% chemically defined lipid concentrate (CDLC) (Gibco), 15 μ g/ml Apo-transferrin, and 450 μ M Mono-Thioglycerol (Sigma) with 7 μ g/ml insulin (Sigma), 10 μ M of SB431542 (Selleckchem), 50 ng/ml Noggin (Peprotech), 1.7 μ M CHIR (Reprocell), and 20 μ M Y-27632 ROCK inhibitor (Stem Cell Technologies). On Day 2, the concentration of Noggin was increased to 100 ng/mL by removing 40 μ L of the media and adding 50 μ L containing 150 ng/ml Noggin in addition to the same concentration of the previous morphogens. To initiate the specification of the midbrain/hindbrain boundary, 40 μ l of media was removed and 50 μ l of 200ng/mL FGF8b was added to each well to bring the final concentration in the well to 100ng/mL at Day 4. Although SB431542, CHIR, and Noggin were still added at the same concentration on Day 4, ROCKi was no longer added to promote cell proliferation and aggregate growth. To promote the cerebellar cell fate, on Day 6, 40 μ l of media was removed and 50 μ l of fresh cerebellar differentiation medium (CerDM) I, containing DMEM/F-12 (Corning/Cytiva), 20% Knockout Serum Replacement (KSR) (Gibco), 15 μ g/ml Apo-Transferrin (Sigma), 7 μ g/ml Insulin (Sigma), 2 mM Glutamax (Gibco), 0.1 mM 2-Mercaptoethanol (Gibco), 100 U/ml penicillin and 100 μ g/ml streptomycin (Corning) was added to each well while supplementing the media with the same final concentration of morphogens as Day 4. On Day 8, the final concentration of FGF8b was increased from 100ng/ml to 300ng/ml to push further toward the midbrain/hindbrain specification and 100 μ l of media was added to each well after the removal of 40 μ l taking the final volume to 150 μ l. On Day 10, 80 μ l of media was removed and supplemented with 100 μ l of media with the same final concentration of the drugs from Day 8. On Day 12 and 14, half media change was conducted without any drugs by removing 70 μ l of media and adding 75 μ l of fresh CerDM1.

From Day 16 to Day 30, organoids were cultured in 10-cm culture plates (Falcon) with orbital agitation (10 r.p.m) in CerDM II, containing DMEM/F12 medium (Corning), 1% N₂ (Gibco), 1% B27 (Gibco), 2 mM Glutamax (Gibco), 100 U/ml penicillin and 100 μ g/ml streptomycin (Corning) and full media change was conducted every 3 days. From Day 30 to Day 60, organoids were cultured in CerDM III, consisting of 1:1 DMEM/F12 (Corning) and Neurobasal medium (Gibco), 1% N₂, 1% B27, 2 mM Glutamax, 5 μ g/ml Heparin (Sigma), 1% Chemically Defined Lipid Concentrate (Gibco), 100 U/ml penicillin, 100 μ g/ml streptomycin (Corning), 0.25 μ g/ml AmphoB (Gibco), 0.5 ng/ml T3 (Sigma) and 1% Matrigel (Corning). From day 60 onwards, organoids were cultured in CerDM IV, consisting of CerDM III without Matrigel and supplemented with 14 ng/ml Brain Derived Neurotrophic Factor (BDNF) (R&D Systems). Full media change was conducted once a week on organoids after day 30.

The differentiation protocols for forebrain, midbrain, and spinal cord organoids used in regional identity qPCR experiments were adapted from previous publications^{60,61}.

Quantitative PCR—Organoids were lysed and the RNA was collected using the RNeasy Mini kit (Qiagen, #74004). Template cDNA was prepared by reverse transcription using the RevertAid First Strand cDNA Synthesis Kit (Thermo Fisher Scientific, #K1621). qPCR was performed using the SYBR Green PCR Master Mix (Thermo Fisher Scientific, #4309155) on a ViiA7 Real-Time PCR System (Thermo Fisher Scientific). Primers used in this study are listed in Mendeley Data Table 4.

Immunohistochemistry—Organoids were fixed in 4% paraformaldehyde for 30 min at room temperature before overnight incubation at 4C in 30% sucrose solution. Organoids were then embedded in Tissue-Tek O.C.T. compound (Sakura, #62550) and cryosectioned at 20µm thickness onto glass slides (Globe Scientific, #1354W). Slides were washed 3x with a 0.1% Tween20 (Sigma #P9416) solution before a 1-hour incubation in 0.3% TritonX-100 (Sigma, #T9284) and 6% bovine serum albumin (Sigma, #AA0281) solution. Slides were incubated for 1-hour at room temperature in a primary antibody solution followed by a 1-hour room temperature incubation in a secondary antibody solution, both consisting of 0.1% TritonX-100 and 2.5% BSA with 3 washes before and after secondary antibody incubation. Slides were coverslipped using Fluoromount G (EMS, #50-259-73). Primary antibodies and dilutions used are specified in Key Resource Table.

Whole organoid staining—Organoids were fixed in 4% paraformaldehyde for 30 min at room temperature before a 1-hour incubation in 0.3% TritonX-100 (Sigma, #T9284) and 6% bovine serum albumin (BSA) (Sigma, #AA0281) solution at 4C with gentle movement. Organoids were then incubated in primary antibody solution for 48 hours at 4C with gentle movement followed by a 48-hour incubation in a secondary antibody solution for 48 hours at 4C with gentle movement, both consisting of 0.1% TritonX-100 and 2.5% BSA with several rounds of washing over a period of 8 hours in a 0.1% Tween20 (Sigma #P9416) solution before and after secondary antibody incubation.

Microscopy and image analysis—Organoids in culture were imaged using a Leica DMIL LED microscope (Leica). Immunofluorescence images were taken with a Leica Model TL LED Thunder widefield scope (Leica) and analyzed using ImageJ⁶² software and MATLAB software. Whole organoids were imaged using a SP8-8X microscope and processed with Leica LASX software.

Calcium Imaging—Organoids were transduced with pAAV-CAG-SomaGCaMP6f⁶³ (Addgene, #158757) using 1.5 µl of virus in 150 µl of CerDM3 medium overnight on static condition. One day after infection, a full media change was performed using CerDM3 media. A full media change to BrainPhys Medium (STEMCELL Technologies, #05792) was performed on day three after transduction. Organoids were kept for at least one week in culture before the imaging session. Organoids were randomly selected and transferred to a recording chamber kept at 37 °C using a heating platform and a controller (TC-324C, Warner Instruments) in BrainPhys Optimized Medium (STEMCELL Technologies, #05796). Imaging was performed using a SP-8X microscope with a multiphoton laser. Time-lapse

images were acquired at 1 frame for 860 ms, using a 25x 0.95 NA water objective (2.5 mm WD) and resulting in a view of 200 x 200 μm^2 . All imaging conditions including excitation light intensity, camera sensor gain, and exposure time were identical for all calcium imaging experiments.

Basal activity was recorded for 10 mins of the imaged organoid. Pharmacological treatment was performed with a bath application of Tetrodotoxin, TTX (Tocris, #1078/1), at a final concentration of 1 μM , 3 μM NMDA (Tocris, #0114) and 3 μM PTX (Hello Bio, #HB0506). Raw tiff calcium imaging files were analyzed using the CNMF and CalMAN package as previously described (CalMAN an open-source tool for scalable calcium imaging data analysis⁴⁹, to identify fluorescent transients and spike estimation in MATLAB (MathWorks). Calcium traces were plotted as relative scaled height in function of time. Hierarchical clustering and pairwise correlation was performed as previously described⁶⁴ with Correlation, linkage parameter was set to 1.5.

Optogenetics and 2P imaging—A fragment of the L7 promoter (854 bp)⁶⁵ was cloned inside of the FUGW lentiviral backbone. The silencing opsin eOPN3 fused to rfp⁵¹ was cloned downstream from the L7 promoter in the lentiviral vector, named L7 eOPN3. Viral particles were produced and concentrated by ultracentrifugation essentially as described in Lois 2002, *Development*⁶⁴. 6-month-old hCerO were transduced with L7-eOPN3 Lentivirus and pAAV-CAG-SomaGCaMP6f using 1.5 μl of each virus into 150 μl of CerDM3 medium overnight in static conditions. Control organoids from the same differentiations were transduced at the same time using only pAAV-CAG-SomaGCaMP6f. Media changes and organoid selection were performed as previously described for SomaGCaMP6f transduction. To activate eOPN3 we used a 525 nm LED stimulation for 500ms at maximal intensity (as in Mahn 2021, *Neurons*⁵¹). As previously described⁵¹ eOPN3 activation lasts for 5 minutes after stimulation, to monitor calcium activity we performed 2P-imaging recordings at 880 nm for 5 minutes immediately after eOPN3 activation. It is important to note that the light intensity used to image gcamp under the 2P microscope is not sufficient to activate ePON3. ePON3 is only activated with a much stronger light intensity that cannot be provided with a 2P microscope. Thus, to activate the sufficient light intensity required to activate eOPN3 we used an LED at 525nm (10mW/mm²).

Whole-cell patch-clamp recording—Organoids were transduced with AAV8.L7-6.eGFP.WPRE.hBG⁶⁵ (Addgene, #126462) using 1.5 μl of virus in 150 μl of CerDM3 medium overnight on static condition in a single well of a 24 well low attachment plates. One day after transduction, a full media change was performed using CerDM3 media. On day three after transduction organoids were transferred to a 6 well low attachment plate and a full media change to BrainPhys Medium (STEMCELL Technologies, #05792) supplemented with NeuroCult SM1 Neuronal Supplement (STEMCELL Technologies, #05711) was performed. Organoids were kept for at least one week in culture before the recordings. Prior to recording, each organoid was transferred into 35mm petri dishes (Ibidi, #80136) on a 10 μl geltrex drop (Thermo Fisher Scientific, #A1413301) and left for 15 mins in a 37°C incubator. Afterwards, organoids were incubated in BrainPhys Optimized Medium (STEMCELL Technologies, #05796) until recording. PCP/L7+ neurons were visualized

under a fluorescence microscope (Olympus BX51 WI). Recordings were performed at RT. Bath application of Tetrodotoxin, TTX (Tocris, #1078/1), at a final concentration of 1 μ M was performed to assess presence of TTX-sensitive APs in Current-clamp mode. Multi-clamp 700B (Molecular Devices) was used for recordings and signals were acquired at 10 kHz using pClamp10 software and filtered at 2 kHz for voltage clamp recordings. Data acquisition was performed with Digidata 1440A (Molecular Devices). Borosilicate glass capillaries were used for patch pipettes ranging between 4.5-8 MOhm. Patch pipettes were filled with intracellular solution (in mM): 122.5 potassium gluconate, 12.5 KCl, 0.2 EGTA, 10 Hepes, 2 MgATP, 0.3 Na3GTP and 8 NaCl adjusted to pH 7.3 with KOH. Intracellular solution was kept on ice during recordings. Passive properties of the membrane were monitored immediately after break-in in current clamp mode. Membrane potential was kept at -70 mV and currents were injected from -100 pA to + 100 pA with 10 pA increments to induce action potentials. Inward sodium and delayed rectifying potassium currents were measured in voltage clamp at depolarizing steps of 10mV. Recording data were analyzed by using pClamp 11 software (Molecular Devices). Baseline gap free signals acquired in current mode were manually adjusted and frequency calculated with automated event detection with event rejection set to 1 ms and 67.26 dV. Gap free traces acquired in voltage clamp mode were used to monitor postsynaptic currents and filtered with a low-pass 2000 Hz Gaussian filter during post hoc traces analysis.

Dissociation of cerebellar organoids and single-cell RNA-seq—Individual brain organoids were dissociated into single cells using the Worthington Papain Dissociation System kit (Worthington Biochemical). To give an estimated recovery of 6,000 cells per channel dissociated cells were resuspended in ice-cold PBS containing 0.04% BSA at a concentration of 1,000 cells/ μ l, loaded onto a Chromium Single Cell 3' Chip (10x Genomics) and processed through the Chromium controller to generate single-cell gel beads in emulsion (GEMs). scRNA-seq libraries were prepared with the Chromium Single Cell 3' Library & Gel Bead Kit v.2 (10x Genomics). Libraries from different samples were pooled based on molar concentrations and sequenced on a NovSeq S1 instrument (Illumina) with 29 bases for read 1, 99 bases for read 2 and 10 bases for index 1 & 2.

Single-cell RNA-seq data analysis—Bioinformatic analysis was performed using the Seurat R package. Briefly, all single cell transcriptomes from 3 organoids were combined, and a set of highly variable genes was determined. The initial PCA analysis was performed using this gene set and the number of statistically significant principal components (PCA) was identified via Jackstraw analysis. Cluster structures were remapped to two dimensions using Uniform Manifold Approximation and Projection (UMAP). Finally, a graph-based clustering approach was used to cluster the cells to produce putative cell-type clusters and return sets of differentially expressed marker genes for each cluster. These were compared to known marker genes to ascribe a cellular identity to each cluster. This data was then integrated with the human dataset to confirm cell type identifications.

Detailed explanation of data integration and analysis—For co-clustering of our cerebellar organoids with human fetal samples, we merged the organoid data set with a downsampled fetal data set using the Seurat pipeline based on Canonical Correlation

Analysis (CCA) integration. First, to downsample the fetal data set, we randomly took 500 cells from each of 21 cell types identified in the fetal data set, (01-PC to 21-BS Choroid/Ependymal). Next, we used the SCTransformation function to both normalize and regress out cycling cells based on the difference between the G2M and S phase scores. We recovered features for integration with the SelectIntegrationFeatures function (features = 3000 genes) and PrepSCTIntegration. Finally, we found the anchoring features with the FindIntegrationAnchors function using normalization.method = "SCT" and anchor.features = features, and integrated the two datasets with the IntegrateData set function.

For the remainder of the analysis, we performed the standard Seurat pipeline to generate PCA and UMAP coordinates before using the FindNeighbors and FindClusters functions (resolution = 0.2, 0.8 and 1.2), to define cell-type clusters. Each cell-type cluster was classified based on known markers and co-clustering with the fetal data.

To calculate the Markers genes, we used the FindMarker function with the test.use attribute set to the default Wilcoxon rank sum test.

Spatial transcriptomics

seqFISH: RNA expression data from multiple sections from two 6-month-old organoids were obtained on the Spatial Genomics platform. Briefly, sections were hybridized overnight at 37 °C with probes against a set of RNA transcripts for Purkinje cell and other cell type specific markers. Samples were washed the following day and loaded onto the automated Spatial Genomics microscope for imaging. Threshold values were manually chosen to select transcript dots with signal above background. Cells were segmented based on expanding the nuclear DAPI stain. Cell-by-gene count matrices were then used for further downstream analyses. Raw counts of transcripts were used for all analyses.

Single-cell pre-processing: The cell-by-gene matrix of counts was pre-processed using the scanpy pre-processing module. For sample 1, cells with less than 1 or greater than 1500 transcript counts were removed. For sample 2, cells with less than 30 or greater than 1500 transcript counts were removed; the higher minimum count threshold was used to remove likely necrotic cells in the center of the organoid.

Identification of Purkinje Cells: Purkinje cells (PCs) were identified using CA8 expression. Cells with CA8 counts greater than or equal to the 99th quantile were defined as PCs; cells with zero CA8 counts were defined as non-PCs and cells with CA8 counts between zero and the 99th quantile were defined as ambiguous.

Differential expression between Purkinje and non-Purkinje cells: We tested for differential expression of gene transcripts between Purkinje cells and non-Purkinje cells as identified above. Ambiguous cells were removed. To avoid psuedo-replication we aggregated cellular transcript counts within samples prior to fitting a gamma-Poisson generalized linear model with glmGamPoi⁶⁶. We accounted for the difference in cell size between Purkinje cells and non-Purkinje cells by dividing cell counts by cell area prior to aggregation. Total transcript counts are often included as a size factor in RNA-seq based psuedo-bulk analysis. This accounts for variation in sequencing depth and cell count

per sample. However, Spatial Genomics transcript counts are image based and therefore transcripts are not resampled. Size factors for the model were the number of cells per pseudo-bulked sample.

QUANTIFICATION AND STATISTICAL ANALYSIS

Data is shown as mean $\pm$ SEM. Statistical analysis was performed using the Graph Pad Prism 6.0. Shapiro-Wilk test was performed to determine the normality of the data ($\alpha=0.05$). Comparisons of means in 2 or more groups were made using an unpaired Student's t-test or analysis of variance (ANOVA). Significant main effects were analyzed further by post hoc comparisons of means using Bonferroni's multiple comparisons test.

SDF1a Layering Quantification—The center of the organoid was initially identified and then divided it up into 3 equal pieces. The bins were placed in the center of each of these pieces on the edge of the organoid.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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HIGHLIGHTS:

- Human cerebellar organoids (hCerOs) develop the fetal cerebellum's cellular diversity
- hCerOs transiently display laminar layering reminiscent of the cerebellar anlage
- Excitatory and inhibitory neurons contribute to network activity in hCerOs
- Purkinje cells in hCerOs display functional features of their in vivo counterparts

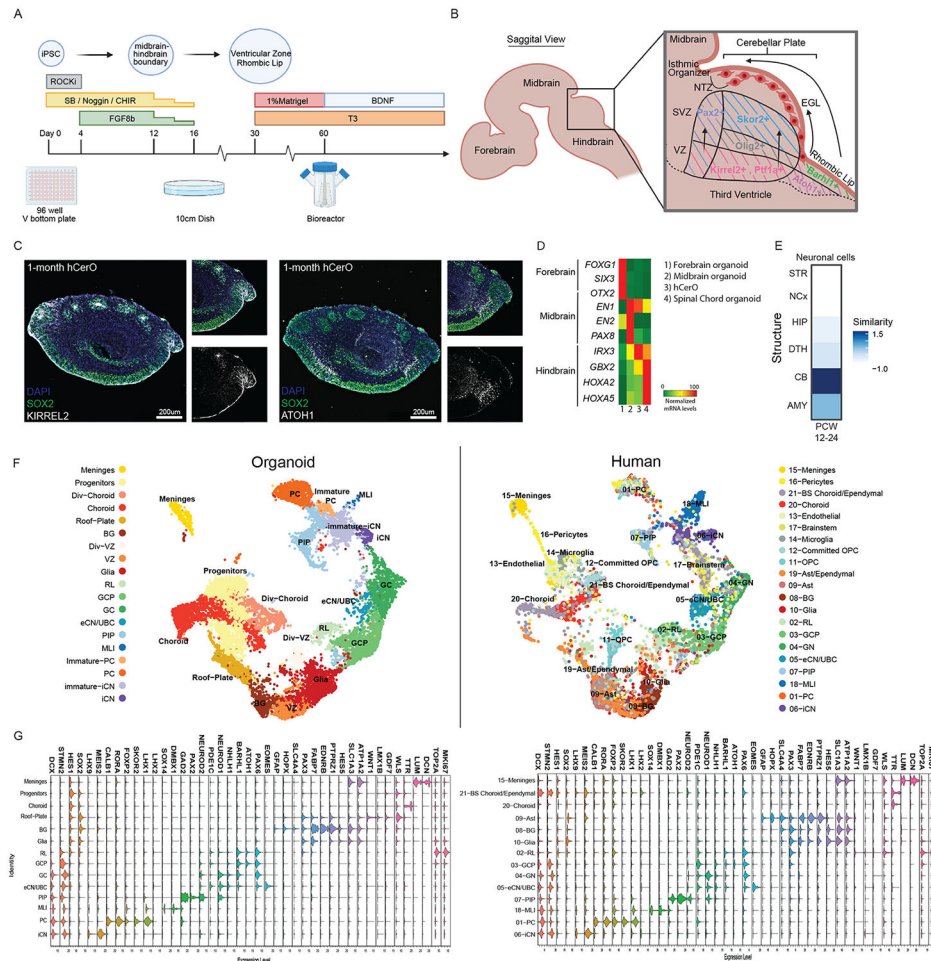


Fig. 1. Human cerebellar organoids (hCerOs) reproducibly generate the cellular diversity of the human cerebellum.

(A) Protocol schematic for generating cerebellar organoids; ROCKi: ROCK inhibition, SB: TGF β inhibition, Noggin: BMP inhibitor, CHIR (CHIR-99021 - GSK3 inhibitor), FGF8b: Fibroblast Growth Factors 8b, T3: triiodothyronine. **(B)** Schematic of the developing human cerebellar plate (sagittal view) EGL: external granular layer, isth: isthmus organizer, mb: midbrain, NTZ: nuclear transitory zone, r1: rhombomere 1, RL: rhombic lip, SVZ: subventricular zone, VZ: ventricular zone. **(C)** Immunofluorescence staining of KIRREL2+ ventricular zone progenitors and ATOH1+ rhombic lip progenitors on 1 month old hCerO. SOX2 defines early neuroepithelium. **(D)** Normalized mean qRT-PCR expression for region-specific markers among various brain region specific organoids at Day 16. **(E)** VoxHunt analysis on neuronal subtypes (GCP, GC, eCN/UBC, iCN, PIP, immature PC, PC, MLI) for PCW 12-24. STR: striatum, NCx: neocortex, HIP: hippocampus, DTH: dorsal thalamus, CB: cerebellum, AMY: amygdala. **(F)** Uniform Manifold Approximation and Projection (UMAP) plot of scRNA-seq data from 2-month organoids after canonical correlation analysis (CCA) batch correction and alignment (D2: one batch; n = 3 individual organoids combined). Left UMAP: combined organoids, colored by cell types. Right UMAP: human dataset. Ast: astrocyte, BG: Bergmann glia, BS: brainstem, CHR: choroid plexus, Div-: dividing, eCN/Unibush: excitatory cerebellar nuclei and unipolar brush cells, GC: granule

cells, GCP: granule cell progenitors, iCN: inhibitory cerebellar nuclei, MEN: meninges, MLI: molecular layer interneurons, OPC: oligodendrocyte progenitor cells, PC: Purkinje cells, PGC: progenitor cells, PIP: pax2+ interneuron progenitors, RG: radial glia, RL: rhombic lip, VZ: ventricular zone. **(G)** Violin Plot of key markers that identify major cell types of the developing cerebellum in organoids vs human fetal tissue.

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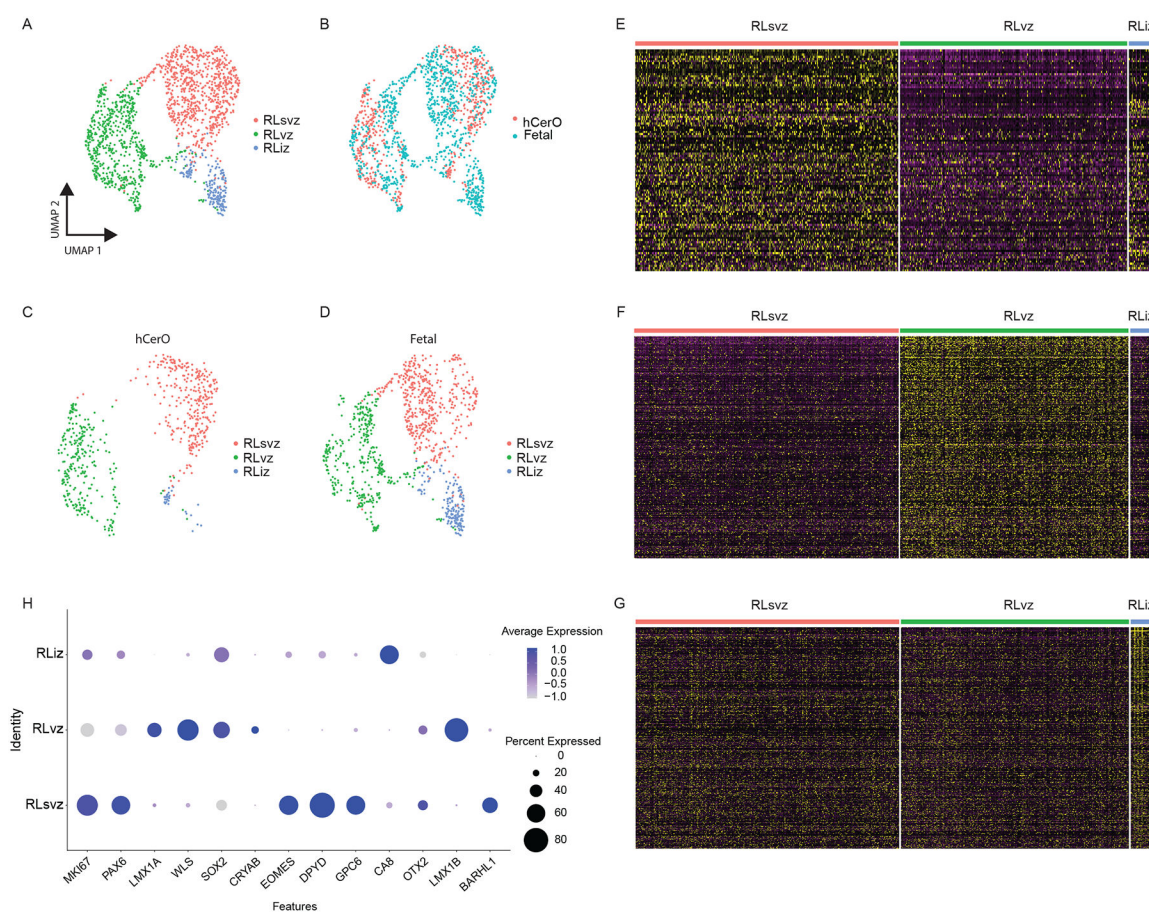


Fig. 2. Identification of human specific rhombic lip ventricular and subventricular subsets. (A) UMAP unbiased clustering and DEG-based annotation of the RL and dividing-VZ subcluster (n=1629 cells; n=788 for SVZ; n=645 for VZ, n=196 for IZ). IZ, intermediate zone; SVZ, subventricular zone; VZ, ventricular zone. (B) UMAP visualization of cells grouped by sample (n=611 for D2 and n=1018 for fetal) (D2 vs fetal). (C, D) UMAP visualization of cells split by sample (D2 vs fetal). (E-G) Heatmap showing the organoid cells' expression of human single cell DEGs from Aldinger et al 2022 used to identify IZ, SVZ, and VZ. (H) Dot plot showing the expression of selected marker genes in each subcluster used in Aldinger et al 2022.

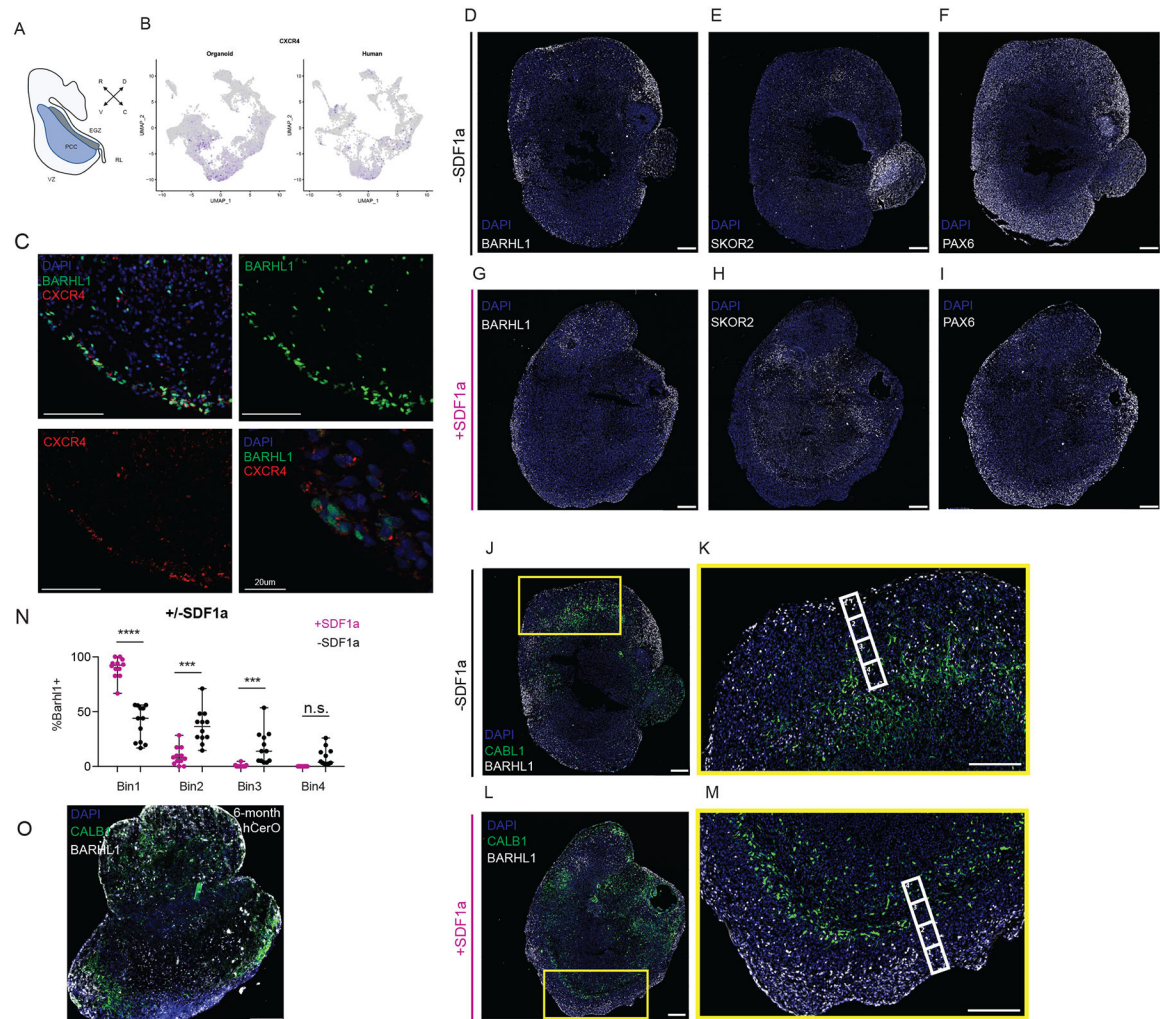


Fig. 3. hCerOs display organized laminar layering reminiscent of the external granule cell layer (EGL) and the Purkinje Cell layer (PCL).

(A) Schematic representation of in vivo human development around Carnegie Stage 23 (CS23) [56 days post conception]. (B) Feature plot of CXCR4 (receptor for SDF1a) within 2-month-old hCerOs and human dataset. (C) Immunofluorescence of hCerOs BARHL1 (granule cell progenitors) and CXCR4 (SDF1a receptor). scale bar: 200um. (D-I) Immunofluorescence of hCerOs BARHL1, SKOR2, and PAX6 (excitatory granule cell progenitor and marker) D-F (-SDF1a) and G-I (+SDF1a) are consecutive sections. (J-M) Immunofluorescence of hCerOs BARHL1 (granule cell progenitors) and CALB1 (Purkinje cells) (N) Binning quantification of BARHL1+ cells. Student's t-test was performed on total of 12 regions for each condition from 3 independent experiments. (O) 6-month-old hCerO no longer treated with SDF1a. ***p-value<0.001, ****p-value<0.0001. scale bar: 200um.

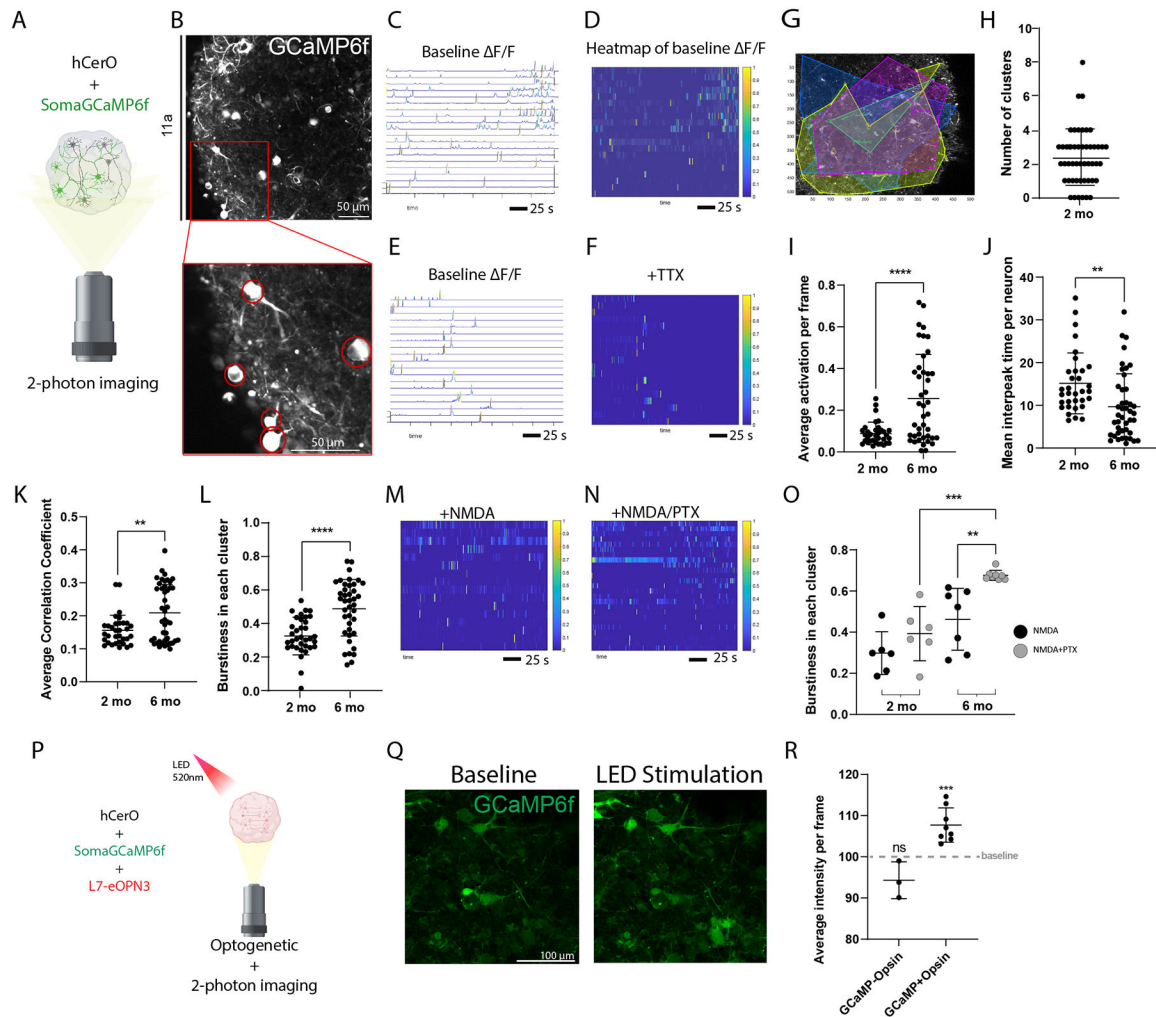


Fig. 4. hCerOs display functionally mature network activity in long-term cultures, resembling patterns of *in vivo* cerebellar circuits.

(A) Schematic showing functional characterization of 2- and 6-month-old hCerOs. (B) Representative 2-month-old 11a organoid transduced with SomaGCaMP6f2. (C) Spontaneous calcium signal traces as $\Delta F/F$ in 2-month-old 11a-derived hCerOs. (D) Spontaneous calcium signal traces as pseudocolor heatmap. (E) Representative heatmap as $\Delta F/F$ after bath application of 1 μ M TTX of a 2-months-old 11a organoid. (F) Representative pseudocolor heatmap after bath application of 1 μ M TTX of a 2-months-old 11a organoid. (G) Representative clustering identification on a 2-month-old hCerO. (H) Quantification of number of clusters in n=48 2-month-old hCerOs. Dots represent average value for each recorded field; bars: standard deviation. (I) Quantification of average activation for frames for n=14 2-months and n=14 6-month-old hCerOs. Student unpaired t test ****p-value<0.0001. Dots represent max average value for each recorded field; bars: standard deviation. (J) Quantification of mean interpeak times for n=14 2-month and n=14 6-month-old hCerOs. Student unpaired t test **p-value=0.0022. Dots represent average value for each recorded field; bars: standard deviation. (K) Quantification of correlation coefficient of calcium events in n=14 2-month and n=14 6-month-old hCerO. Student unpaired t test

******p-value=0.0020. Dots represent average value for each recorded field; bars: standard deviation. **(L)** Quantification of burstiness in each cluster for n=14 2-month and n=14 6-month-old hCerO. Student unpaired t test *******p-value<0.0001. Dots represent average value for each recorded field; bars: standard deviation. **(M)** Representative heatmap after bath application of 3 μ M NMDA of a 2-months-old 11a organoid. **(N)** Representative heatmap after bath application of 3 μ M NMDA and 3 μ M PTX of a 2-months-old 11a organoid. **(O)** One way ANOVA comparison of burstiness in each cluster of n=6 2-month-old and n=7 6-month-old hCerOs treated with NMDA and NMDA+PTX. Dots represent average value for each recorded organoid; bars: standard deviation. *******p-value=0.005, ******p-value=0.0065. **(P)** Schematic representation of 6-months-old hCerO transduction with AAV8-SomaGCaMP6f and Lentivirus L7-eOPN3. Optogenetic stimulation was performed with 520 nm LED. Calcium imaging was performed with 2-photon microscopy. **(Q)** Representative images of maximum projection of stacks acquired during calcium imaging recording. **(R)** Analysis of average intensity per frame from calcium imaging analysis in n=3 eOPN3⁻ and n=8 eOPN3⁺ infected organoids. Each dot represents an organoid. Student paired t-test *******p-value=0.0001.

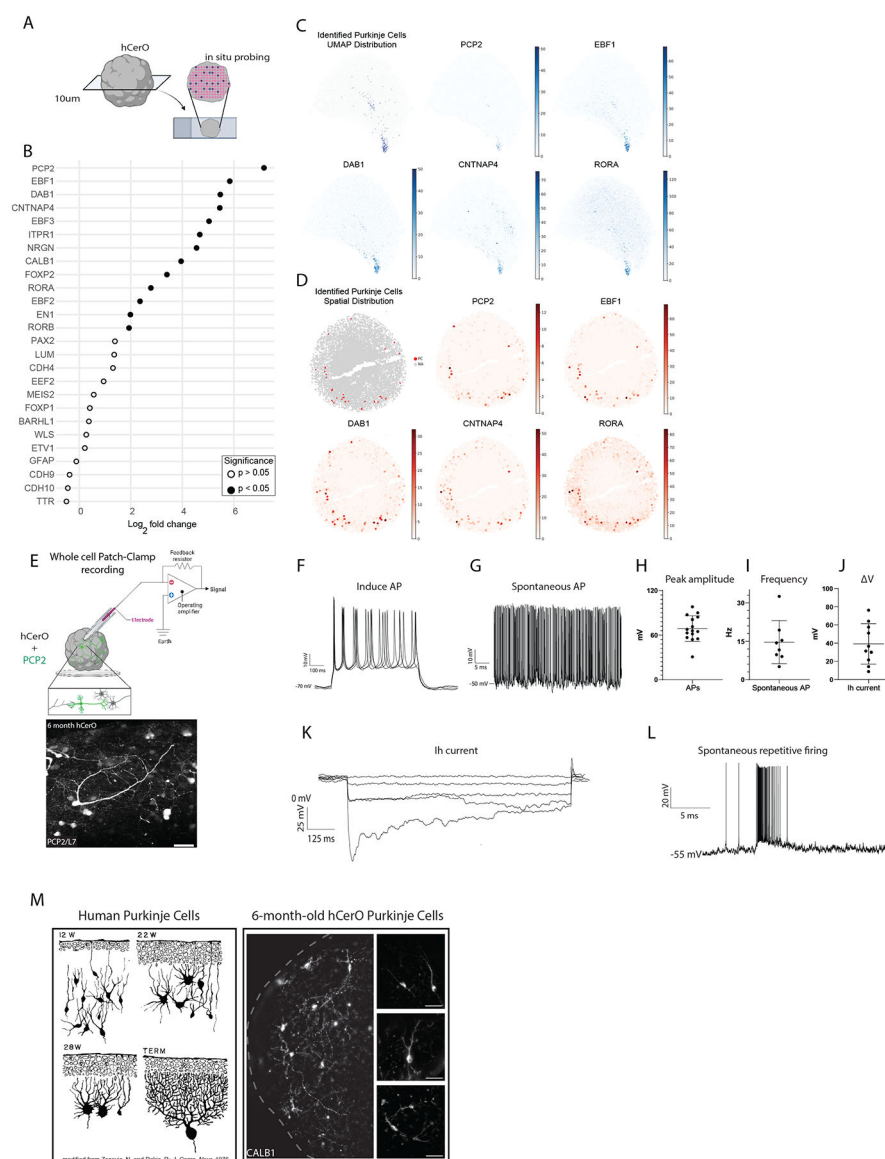


Fig. 5. Functionally mature, human Purkinje neurons develop within long-term culture of hCerOs.

(A) Schematic representation of spatial transcriptomics conducted on hCerOs. (B) Differential expression of genes between cells identified as Purkinje and non-Purkinje cells in 6-month-old hCerOs. Significant changes (p-value<0.05) are marked with solid circle. (C) UMAP feature plots showing cell level expression of Purkinje cell marker genes identified with overlapping expression of high CA8 levels. (D) Spatial localization of upregulated Purkinje cell genes within a section of 6-month-old hCerO sample #2. (E) Upper panel: schematic of patch clamp recording of PCP2+ neurons from intact hCerOs. Lower panel: representative PCP2+ neuron. Imaged with SP8-8X microscope. (F) Representative whole-cell patch-clamp trace of induced AP from PCP2+ neuron from intact hCerOs. (G) Representative whole-cell patch-clamp trace of spontaneous firing from PCP2+ neuron from intact hCerOs. (H) Quantification of peak amplitude for AP elicited from n=14 recorded cells. Mean + SD. Each dot represents a single AP from an individual neuron. (I) Quantification of frequency for spontaneous AP. (J) Quantification of ΔV for Ih current. (K) Representative whole-cell patch-clamp trace of Ih current. (L) Representative whole-cell patch-clamp trace of spontaneous repetitive firing. (M) Human Purkinje Cells and 6-month-old hCerO Purkinje Cells.

(I) Quantification of AP frequency from 8 spontaneously firing cell. Mean + SD. Each dot represents a recorded neuron. **(J)** Quantification of I_h current for n=10 Inward Sodium/Outward Potassium Rectified currents. Mean + SD. Each dot represents a recorded neuron. **(K)** Representative whole-cell patch-clamp trace of I_h current from PCP2+ neuron from intact hCeOs. **(L)** Representative whole-cell patch-clamp trace of repetitive spontaneous firing from PCP2+ neuron from intact hCerOs. **(M)** Drawings of human Purkinje cells at various developmental stages⁵⁴ (Figure reprinted with permission from Zecevic et al., 1976) compared to 6-month-old hCerO stained for CALB1. Imaged with Leica Model TL LED Thunder widefield scope.

Key resources table

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Rabbit polyclonal anti-BARHL1	Proteintech	CA#HPA004809
Rabbit polyclonal anti-SKOR2	Proteintech	CA#HPA046206
Rabbit polyclonal anti-KIRREL2	Proteintech	CA#10890-1-AP
Goat polyclonal anti-SOX2	RD systems	CA#AF2018
Rabbit monoclonal anti-ATOH1	Abclonal	CA#A11477
Mouse monoclonal anti-CALB1	Sigma-Aldrich	CA#C9848-100UL
Rabbit polyclonal anti-PAX6	Biolegend	CA#901301
Chemicals, peptides, and recombinant proteins		
Accutase	Sigma	07922
AmphoB	Gibco	15290018
Apo-transferrin	Sigma	T1147-100MG
BDNF	R&D Systems	248-BDB-050
BrainPhys Optimized Medium	STEMCELL Technologies	05796
Bovine Serum Albumin (BSA)	Sigma	A0281
B-27 supplement	Gibco	17504001
chemically defined lipid concentrate (CDLC)	Gibco	11905031
CHIR	Reprocell	04-0004
DMEM/F-12	Corning/Cytiva	SH30023.FS
Fluoromount	EMS	50-259-73
F-12 medium	Gibco	11-765-054
Geltrex	Thermo Fisher Scientific	A1413301
Glutamax	Gibco	35050061
Heparin	Sigma	H314-25KU
IMDM medium	Gibco	12440053
insulin	Sigma	I9278-5ML
Knockout Serum Replacement (KSR)	Gibco	10828028
Matrigel	Corning	354230
Mono-Thioglycerol	Sigma	M1753
mTeSR1medium	Stem Cell Technologies	4309155
Neurobasal medium	Gibco	21103049
NeuroCult SM1 Neuronal Supplement	STEMCELL Technologies	05711
NMDA	Tocris	0114
Noggin	Peprotech	120-10C
N2 supplement	Gibco	17502001
Penicillin/streptomycin	Corning	SV30010
PTX	Hello Bio	HB0506

REAGENT or RESOURCE	SOURCE	IDENTIFIER
ROCK inhibitor	STEMCELL Technologies	72302
SB431542	Selleckchem	S1067
SYBR Green PCR Master Mix	Thermo Fisher Scientific	4309155
Tetradotoxin, TTX	Tocris	1078
Tissue-Tek O.C.T. compound	Sakura	62550
TritonX-100	Sigma	T9284
Tween20	Sigma	P9416
T3	Sigma	T2877-250MG
2-Mercaptoethanol	Gibco	21985023
Critical commercial assays		
RNeasy Mini kit	Qiagen	74004
RevertAid First Strand cDNA Synthesis Kit	Thermo Fisher Scientific	K1621
Deposited data		
RNA-seq data of human cerebellar organoids (hCerOs)	This paper	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE247974
Atamian_et_al_2023_deposited figures and tables	This paper; Mendeley Data	https://data.mendeley.com/preview/6hr4sgn4sx?a=1a6b251c-8d1e-4089-95e5-d5fad93a5c78
Supplementary Text	This paper; Mendeley Data	https://data.mendeley.com/preview/6hr4sgn4sx?a=1a6b251c-8d1e-4089-95e5-d5fad93a5c78
RNA-seq data of developing human cerebellum	Aldinger et al. ³⁷	accession number phs001908.v2.p1
Experimental models: Cell lines		
Human: PGP1 (Personal Genome Project 1) hiPSC	Laboratory of Gorge Church	hiPSC
Human: PGP1-FOXG1 hiPSC	Harvard Stem Cell institute	hiPSC
Human: WA09 (H9) hESC (NIH approval number NIHhESC-10-0062)	WiCell	hES Cell Line: H9
Human: 11a hiPSC	Laboratory of Kevin Eggan	hiPSC
Human: D2 hiPSC	Laboratory of Marcelo Coba	hiPSC
Oligonucleotides		
RT-PCR Primer Sequence	See Mendeley Data	Table 4.
Recombinant DNA		
Plasmid: pAAV-CAG-SomaGCMP6f2	Shemesh et al. ⁶³	Addgene catalog #158757
Plasmid: pAAV/L7-6-GFP-WPRE	Nitta et al. ⁶⁵	Addgene catalog #126462
Software and algorithms		
Biorender	https://www.biorender.com/	N/A
Human + Organoid Integrated data code	This paper	https://github.com/quadratolab/Cerebellar-Organoid-2023
ImageJ	Schneider et al. ⁶²	https://imagej.nih.gov/ij/
MATLAB	MathWorks	2022b
Other		
Ultra-low attachment V-bottom Plates	Sbio	MS-9096VZ

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Ultra-low attachment 6-well dishes	Corning	3471
10cm plates	Avantor	10062-880

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